

Exploratory Data Analysis (EDA) Report

Netflix Movies and TV Shows Dataset

ApexPlanet Data Analytics Internship - Task 2

1. Introduction

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Netflix Movies and TV Shows dataset. The analysis aims to uncover patterns, trends, and insights related to Netflix content distribution, ratings, release trends, and genre popularity.

2. Dataset Overview

Attribute Value

Total Records 8807

Total Columns 12

Dataset Type Netflix Movies and TV Shows

Dataset Columns

- show_id
 - type
 - title
 - director
 - cast
 - country
 - date_added
 - release_year
 - rating
 - duration
 - listed_in
 - description
-

3. Data Cleaning Summary

The following preprocessing steps were performed before analysis:

- Handled missing values in director, cast, country, rating, and duration columns.
- Converted date_added to datetime format.
- Verified duplicate records.
- Created a cleaned dataset for analysis.

Missing Values Found

Column	Missing Values
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director	2634
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cast	825
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country	831
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date_added	10
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rating	4
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duration	3
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Duplicate Records

- Duplicate Rows Found: 0

4. Content Type Analysis

Movies vs TV Shows

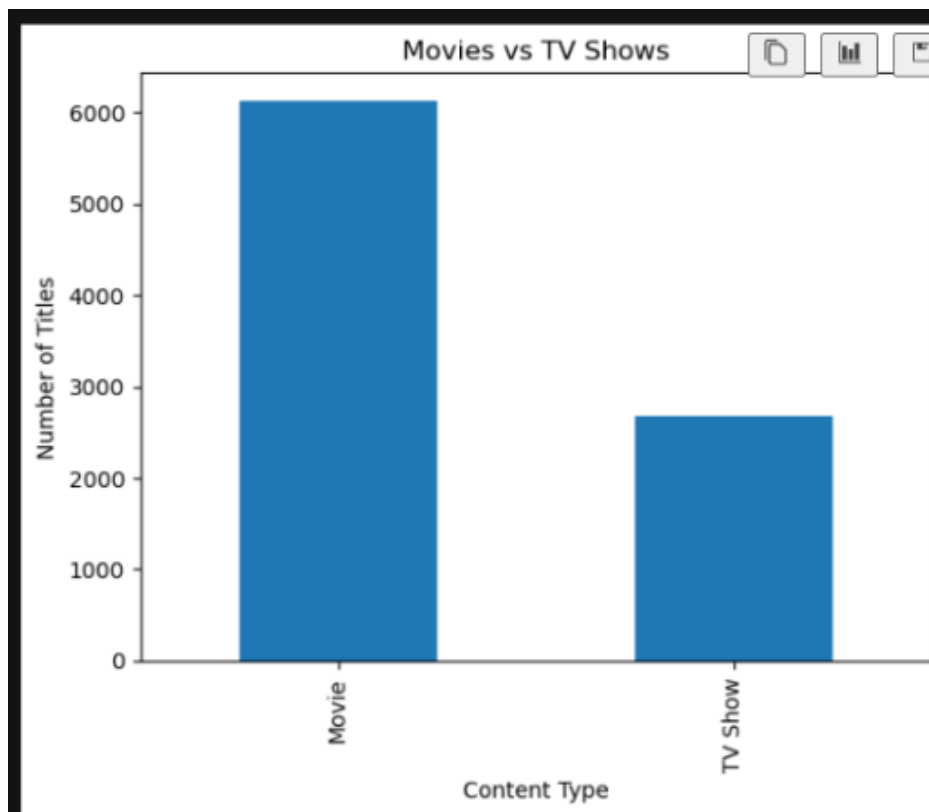
Type	Count
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Movie	6131
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TV Show	2676
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Insight

Movies dominate the Netflix catalog, accounting for approximately 69.6% of all titles, while TV Shows account for approximately 30.4%.



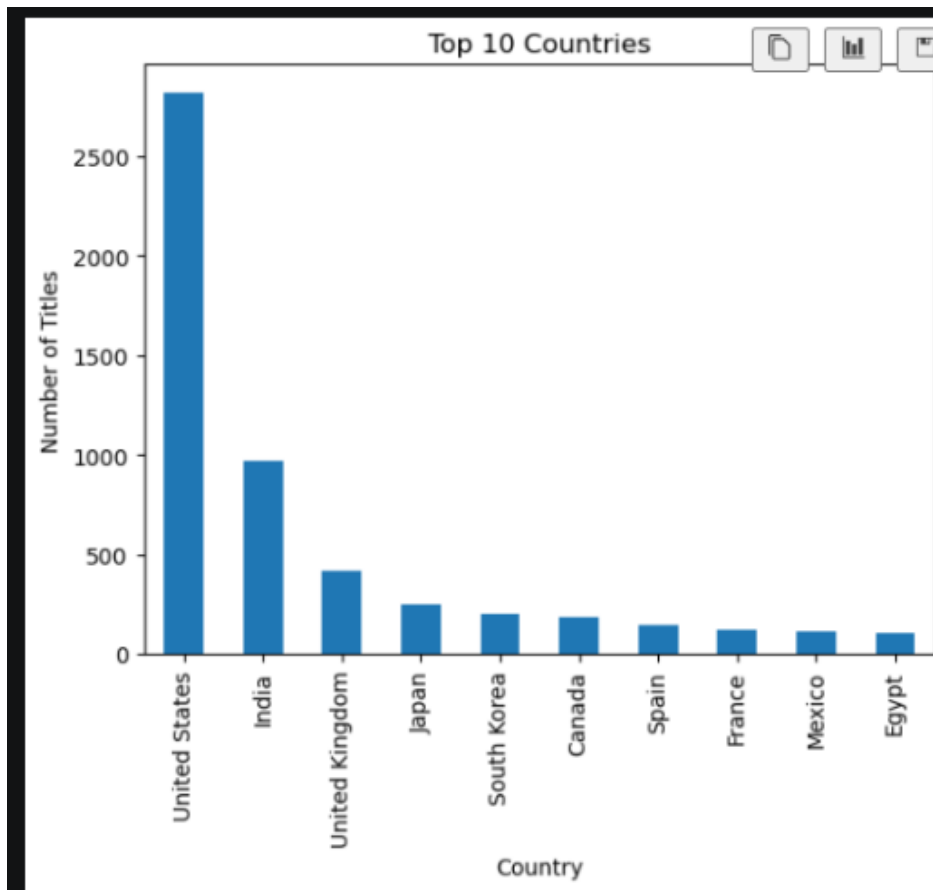
5. Country Analysis

Top Content Producing Countries

Country	Titles
United States	2818
India	972
United Kingdom	419
Japan	245
South Korea	199

Insight

The United States is the largest contributor to Netflix's content library, followed by India and the United Kingdom.



6. Ratings Analysis

Most Common Ratings

Rating Count

TV-MA 3207

TV-14 2160

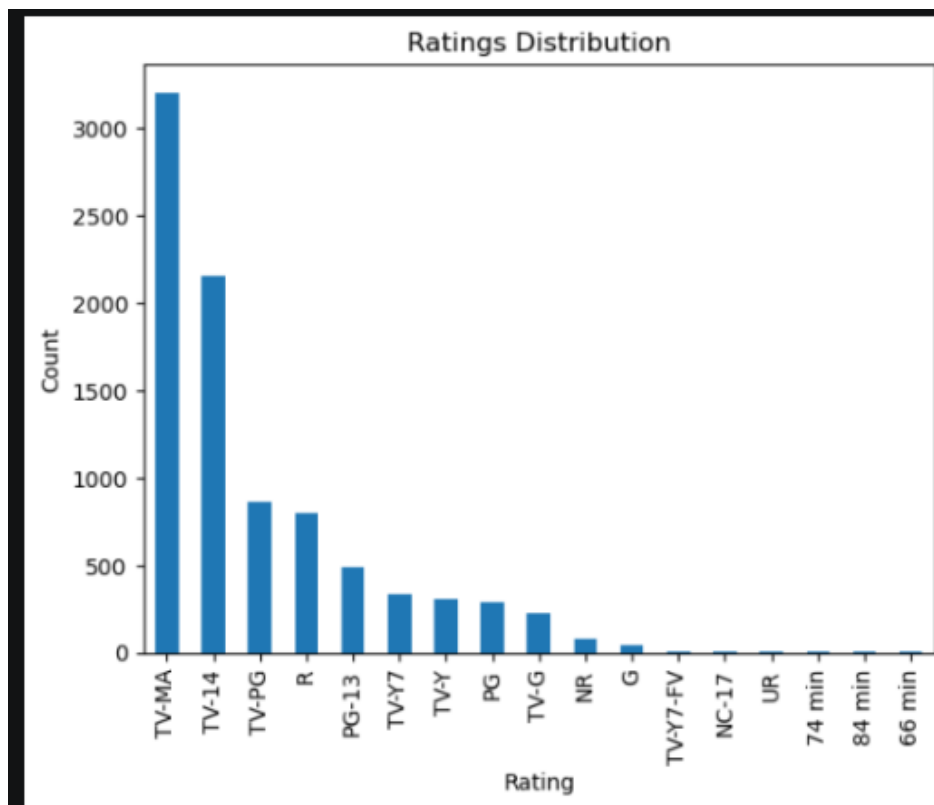
TV-PG 863

Insight

TV-MA is the most common content rating, indicating that Netflix primarily targets mature audiences.

Additional Observation

Three invalid rating values (74 min, 84 min, and 66 min) were identified, suggesting data quality issues in the original dataset.



7. Release Year Analysis

Recent Release Trends

Year Titles

2017 1032

2018 1147

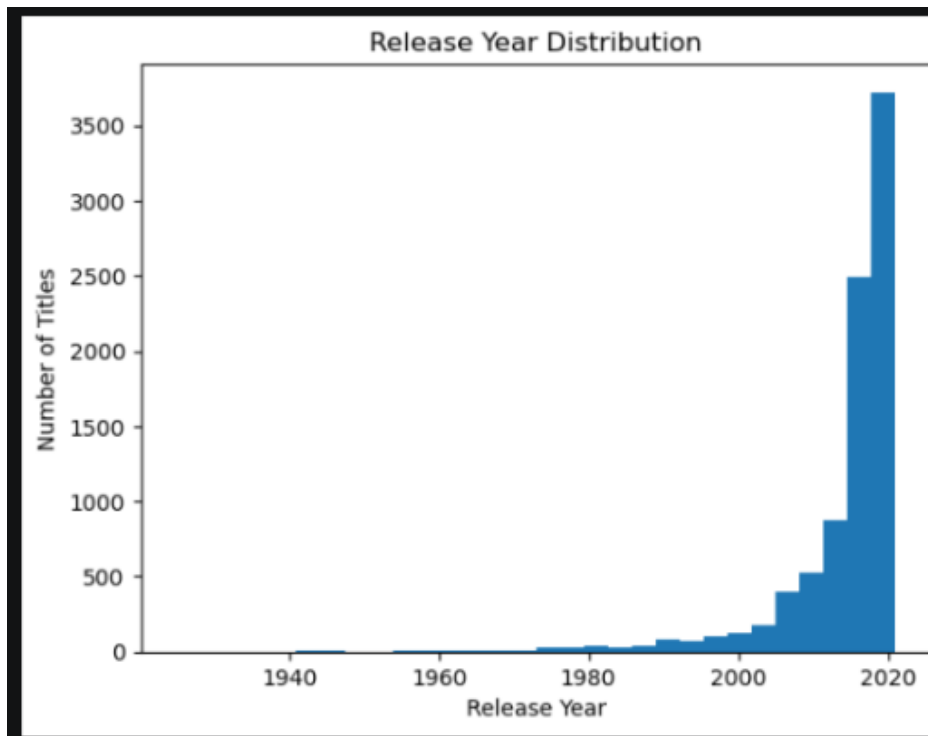
2019 1030

2020 953

2021 592

Insight

Netflix experienced significant content growth between 2012 and 2018, reaching a peak of 1,147 titles in 2018. Content releases declined slightly after 2018.



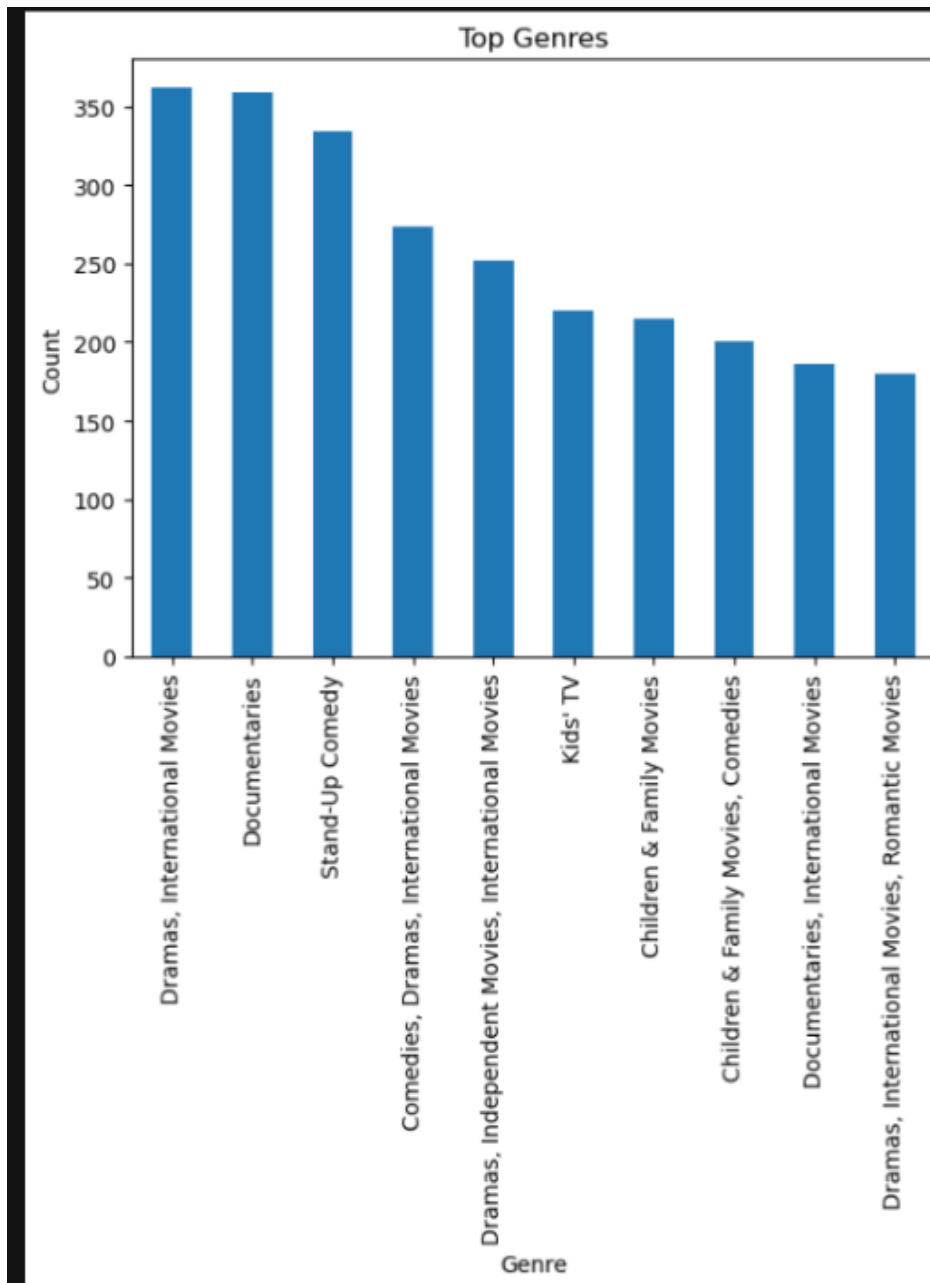
8. Genre Analysis

Top Genres

Genre	Count
Dramas, International Movies	362
Documentaries	359
Stand-Up Comedy	334

Insight

Drama-based content dominates Netflix's catalog. International Movies appear frequently among the top genres, highlighting Netflix's focus on global audiences.



9. Key Business Insights

1. Movies account for nearly 70% of Netflix content.
 2. The United States and India are the leading content producers.
 3. Netflix primarily targets mature audiences through TV-MA and TV-14 content.
 4. Content production peaked in 2018.
 5. Drama and International Movies are the most represented genres.
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10. Conclusion

The Exploratory Data Analysis revealed several important patterns within Netflix's content library. Movies dominate the platform, the United States leads content production, mature audience ratings are most common, and drama-based international content forms a significant portion of the catalog. These insights can support future business intelligence, dashboard development, and strategic decision-making.

Tools Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Jupyter Notebook
- SQL